Rapid Test Kit

Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Rapid test kits for Malaria, Dengue, HIV, HCV, Typhoid, Pregnancy

What it is used for

A rapid test kit detects specific markers in a sample to support screening for selected infections or pregnancy. Results must be interpreted according to the kit instructions and confirmed by appropriate healthcare professionals when required.

Main parts

- Identify the test device, cassette, strip or reaction card supplied with the kit.
- Locate the sample collection device, such as a dropper, swab or lancet, if included.
- Identify the supplied buffer, reagent or extraction solution, where applicable.
- Locate the test buffer wells and result window, if present.
- Identify the control and test result markings specified by the manufacturer.
- Locate the sterile gloves, sample container and biohazard waste materials.
- Identify the package insert, expiry date and batch number.

Before you start

1. Use only an authorised kit appropriate for the intended screening purpose.
2. Read the complete instructions and check storage conditions and expiry date.
3. Wear appropriate PPE and prepare a clean, well-lit testing area.
4. Confirm the required sample type and collection method for the specific kit.
5. Collect and label samples using approved procedures and correct identification practices.
6. Check that all kit components are present and undamaged.
7. For blood-borne infection testing, ensure privacy, informed consent and trained supervision.

How to use

1. Allow the sealed kit and reagents to reach the specified temperature if instructed.
2. Prepare the sample using the collection method specified for that particular test.
3. Open the test device only when ready to use it.
4. Add the specified sample quantity and buffer using the supplied equipment.
5. Start timing according to the manufacturer's instructions.
6. Keep the device level and undisturbed during the reaction period.
7. Read the result only within the specified time window.
8. Check the control indicator to determine whether the test is valid.
9. Record the result confidentially and explain that screening may require confirmatory testing.
10. Dispose of used devices, lancets and contaminated materials in appropriate biomedical waste containers.

⚠ Safety

- Treat all human samples as potentially infectious and wear appropriate PPE.
- Never reuse lancets, needles, swabs or other single-use collection items.
- Do not interpret an invalid test as positive or negative.
- A screening result alone does not establish a definitive diagnosis for infections such as HIV or HCV.
- Follow approved procedures for consent, confidentiality, sample handling and biomedical waste disposal.
- Seek qualified medical guidance for positive, negative or inconclusive results according to the test and clinical situation.

Care & storage

- Store unopened kits within the manufacturer's specified temperature and humidity limits.
- Keep reagents sealed and protect test components from contamination and direct sunlight.
- Record batch numbers, expiry dates and quality-control checks where required.
- Clean the work area and replenish supplies after each testing session.

Common mistakes to avoid

- Using expired kits or kits stored outside specified conditions.
- Adding an incorrect sample volume or the wrong buffer.
- Reading results too early or after the permitted time window.
- Treating a screening result as a final diagnosis without required follow-up.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar